A computer-assisted construction of a $c = 8$ Haagerup CFT

Proof manuscript with explicit review obligations

Research manuscript

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Abstract

We describe an explicit conformal vertex subalgebra of the lattice VOA $V_{E_6 \oplus A_2}$, generated by an $A_{1,1} \otimes A_{1,39}$ current algebra, the ambient stress tensor, and one three-term lattice field. Exact finite certificates, together with a recursive strong-generation argument, give a proposed computer-assisted proof that its current space has dimension six. We present the subsequent regularity, branching, modular-character and categorical arguments leading to a proposed identification with the Haagerup center. The full unitary-CFT conclusion depends on the analytic claims highlighted in red. Additional delicate local deductions and imported proof components are marked in amber. This document records a proof for review; it is not a report of external expert validation or proof-assistant verification.

HIGH PRIORITY REVIEW — Status of the main claim

The computational closure certificates are complete. Their completion does not mechanically verify the new analytic arguments used later in the manuscript. In particular, the net-sector comparison, module reconstruction/fullness, and annulus compatibility require specialist review. The complete-unitarity and unitary ribbon-equivalence conclusions are stated conditionally on those claims below. Earlier research notes that call the entire identification "complete" should be read with this explicit qualification. Reviews by additional agents are internal reviews, not independent human expert validation.

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1 Statement, conventions, and proof dependencies

Let

$$B = V_{E_6 \oplus A_2}, \qquad H = V_{E_8}, \qquad B \subset H$$

be the standard index-three lattice extension. The notation $A_{1,k}$ means the simple affine VOA $L_k(\mathfrak{sl}_2)$. Fix the embedding $\mathcal{K} = A_{1,1} \otimes A_{1,39} \subset B$ described in Section 2. Let $u \in B_2$ be the field in (5), and define the *actual generated* vertex algebra

$$U = \langle \mathcal{K}, \omega_B, u \rangle \subset B. \tag{1}$$

All products are computed in this lattice VOA. Thus associativity is inherited from the ambient algebra, rather than assumed for an abstract presentation inferred from character coefficients. Throughout the normal-space and mode formulas, $K = \mathcal{K}_1$ denotes only the six-dimensional current space, not the full affine VOA.

Proposition 1.1 (Computational properness conclusion)**.** *Accept the exact lattice identities and provenance recorded in Section A. The algebra U is strongly generated by actual fields of weights at most four. Its weight-one space is*

$$U_1 = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2, \qquad \dim U_1 = 6.$$

Consequently $U \subsetneq B$, since $\dim B_1 = 86$.

The finite certificates and the all-weight implication are presented in Sections 3–6. This is the most directly checkable part of the construction.

Claim 1.2 (Full identification proposed by the manuscript)**.** *With the additional local deductions and analytic claims marked for review in Sections 7 onward, U is completely unitary and strongly rational, and*

$$c(U) = 8, \qquad \mathrm{Rep}(U) \simeq Z(\mathcal{H}_3), \tag{2}$$

$$\mathrm{ch}_U(\tau) = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{3}$$

The equivalence is an equivalence of unitary ribbon categories. The full twelve-component character vector is the $\gamma = 0$ Evans–Gannon vector. There is a unitary diagonal full CFT with $c_L = c_R = 8$ and partition function $\sum_i |\chi_i(\tau)|^2$. A choice of holomorphic extension $H' \cong V_{E_8}$ has all-module category $\mathcal{H}_3$, and its algebra-bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$.

Here H' is obtained by transporting the canonical $\mathcal{H}_3$ Lagrangian algebra. It need not coincide with the originally embedded extension H/U; the proposed argument identifies that original all-module category only up to the alternatives $\mathcal{H}_2, \mathcal{H}_3$. The full-CFT defects in Claim 1.2 preserve U, and need not commute with the enlarged E_8 chiral algebra.

1.1 How the qualifications are used

Amber **REVIEW** notes identify deductions whose complete verification is not supplied by the finite closure computations, or whose long local proofs are imported from companion notes. Red **HIGH PRIORITY REVIEW** notes identify claims about which confidence is lower, especially operator domains and the passage from conformal nets to complete VOA unitarity. A highlighted claim is not being declared false. It is deliberately made visible as a premise of the downstream conclusion.

Part of the argument	Nature of its evidence
Explicit fields and finite closure	Exact arithmetic over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$, saved recipes and full residuals.
Properness in all weights	Recursive algebraic theorem plus finite certificates; no character upper-bound extrapolation.
Rationality and branching	Positive-stress and category arguments, finite-top certificates, and cited external theorems.
Haagerup modular data	Rank exclusions, indicator and character arguments; some long derivations remain companion proof inputs.
Complete unitarity and unitary identification	New analytic comparison claims, followed by cited unitary classification and extension theorems.

2 The explicit lattice construction

Use simple lattice vectors indexed $0, \ldots, 7$. The Gram matrix has diagonal entries two and off-diagonal entries minus one on

$$(0,1), (1,2), (2,3), (3,4), (2,5), (6,7),$$

with all remaining off-diagonal entries zero. Fix the cocycle

$$\epsilon(a,b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}. \tag{4}$$

The first current factor corresponds to the root $a_0 = (-1,-2,-3,-2,-1,-2,0,0)$. The second is the diagonal of the principal A_1 in A_5, at level 35, and the principal A_1 in the external A_2, at level four.

The exact coefficient convention uses unnormalized simple-root Heisenberg oscillators and a torus gauge in which the principal raising current is the sum of the corresponding simple-root vertices. The lowering current is $-\sum_i c_i e^{-\alpha_i}$, with

$$(c_0, \ldots, c_7) = (5,8,9,8,5,0,2,2).$$

The torus rescales simple-root vertices by $(1/\sqrt{5}, 1/\sqrt{8}, 1/3, 1/\sqrt{8}, 1/\sqrt{5}, 1, 1/\sqrt{2}, 1/\sqrt{2})$. In this convention define

$$\begin{aligned} u = {} & e^{(0,0,0,0,0,-1,1,1)} \\ & + \frac{\sqrt{13} + 3\sqrt{-3}}{20} e^{(0,0,0,1,1,-1,0,0)} \\ & + \frac{\sqrt{13} - 3\sqrt{-3}}{20} e^{(1,1,0,0,0,-1,0,0)}. \end{aligned} \tag{5}$$

The compact real form and PCT operation must be transported through this gauge. Ordinary coefficient conjugation in the displayed gauge is not a replacement for the ambient PCT map.

Certificate input 2.1 (Actual initial spaces)**.** The saved exact construction gives a PCT-stable 120-dimensional space $S_2 \subset U_2$, containing u, ω_B, the affine vacuum descendants, two neutral primitive fields, and eight charged current multiplets. The current multiplet dimensions are

$$(1,5), (1,9), (1,9), (1,13), (2,2), (2,6), (2,8), (2,14).$$

These contribute 96 vectors, in addition to the 22 affine vacuum vectors and two neutral fields. The exact generated grade-three space $S_3 \subset U_3$ has dimension 956. Its normal subspace $N_3 = K_{-1}S_2 + TS_2$ has dimension 634, and

$$S_3 = N_3 + \operatorname{span}\{a_0 b : a, b \in S_2\}, \qquad \dim(S_3/N_3) = 322.$$

The spaces are current-stable. These are dimensions of constructed finite spaces, not initial assertions that they exhaust U_2, U_3.

The source records are `hi_cft_20260923_c8_exact_fields.json`, `hi_cft_20260923_c8_exact_unitary_certificate.json`, and `hi_cft_20260923_c8_minimal_generators.json`, together with the exact grade-three checkpoint bound into the final ledger.

3 Finite closure and the all-weight criterion

Write $Y(a,z) = \sum_{n\in\mathbb{Z}} a_n z^{-n-1}$ and $T = L(-1)$. A singular product has mode index $n \geq 0$. The zero mode a_0 is a derivation and commutes with T.

Theorem 3.1 (Recursive strong-generation criterion)**.** *Let V be a nonnegatively graded vertex algebra with $V_0 = \mathbb{C}\mathbf{1}$. Choose finite homogeneous subspaces $S_d \subset V_d$, $1 \leq d \leq h$, and let $\mathcal{N}$ be their differential normally ordered span, allowing every parenthesization. Put $K = S_1$ and $A = S_2$. Assume*

1. *$k_n s, a_n s \in \mathcal{N}$ for every $k \in K$, $a \in A$, $s \in S_d$, and $n \geq 0$;*
2. *for $d \geq 3$,*
$$S_d \subset A_0 S_{d-1} + \mathcal{N}_d(S_1 + \cdots + S_{d-1}).$$

Then $\mathcal{N}$ is a vertex subalgebra. In particular, the chosen spaces strongly generate the vertex algebra that they generate.

Proof. The first assumption, the derivation property and $[T, a_0] = 0$ give $a_0\mathcal{N} \subset \mathcal{N}$ for $a \in A$. We first recall the filtered Wick observation. If all generator pairs of total weight less than w have singular products in $\mathcal{N}$, the same is true for normal expressions of total weight less than w. At total weight w, a derivative or a product with at least two positive-weight factors also reduces to smaller total weights. Indeed,

$$[x_\lambda : yz :] = \, : [x_\lambda y]z : + : y[x_\lambda z] : + \int_0^\lambda [[x_\lambda y]_\mu z]\, d\mu.$$

Each omitted positive-weight factor lowers the first two pair weights. In a nested term, $x_j y$ has weight $\operatorname{wt}(x)+\operatorname{wt}(y)-j-1$, so its pairing with z has total weight at most $w-1$. Sesquilinearity handles derivatives and skew symmetry handles a composite on the left. These are identities inside V.

Now induct lexicographically on $(\operatorname{wt}(s) + \operatorname{wt}(t), \operatorname{wt}(s))$ for generator pairs s, t. Left weight one or two is the first assumption. Otherwise the second assumption reduces the left input to $s = a_0 b$, with $\operatorname{wt}(b) = \operatorname{wt}(s) - 1$, plus normal remainders. The latter are covered by the filtered Wick observation. Use

$$(a_0 b)_n t = a_0(b_n t) - b_n(a_0 t). \tag{6}$$

The first term has smaller total pair weight before applying a_0, which preserves $\mathcal{N}$. The known normal expression for $a_0 t$ has weight $\operatorname{wt}(t) + 1$. Every derivative or composite term reduces

total pair weight in the second term. The only remaining possibility is a single undifferentiated generator of weight $\mathrm{wt}(t)+1$; its pairing with b has the original total weight and smaller left weight. It is therefore an earlier induction case. If $\mathrm{wt}(t)=h$, this single-generator possibility is absent.

All generator singular products close. Wick induction then gives closure for all normal expressions. Translation and normal multiplication preserve $\mathcal{N}$ by definition, so all negative modes close too. □

For the Wick identities see De Sole–Kac, Section 2. The full local derivation is `hi_cft_20260923_next_structure_recursive_strong_certificate.md`.

REVIEW — F2: the all-weight implication is an algebraic proof

The filtered Wick reduction and the lexicographic induction above are new deductions connecting finitely many gates to closure in every weight. Their correctness is not verified by checking more low-weight products. In particular, a reviewer should check the reduction of composite normal expressions and the single-generator case at unchanged total weight. The finite certificates below supply the hypotheses; they do not replace this argument.

3.1 The finite spaces and prerequisite controls

Choose actual quotient lifts $P_3 \subset S_3$ and an actual triplet $P_4 \subset U_4$, with joint current highest weight $(0,2)$, and set

$$N_4 = K_{-1}S_3 + TS_3 + (S_2)_{-1}S_2, \qquad W_4 = N_4 + P_4, \tag{7}$$

$$N_5 = K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3. \tag{8}$$

These denote full differential normal spaces, using the lower closure identities to reorder expressions. A computational solver may use a proper subset of actual words in these spaces.

The triplet is obtained from an actual $a_0 p$ by current-zero-mode projectors and lowering. It satisfies $P_4 \subset (S_2)_0 S_3 + N_4$. At ambient grade four, the two Cartan norms $2, 78$ bound highest weights by $4, 24$, respectively. Consequently finite polynomials in FE give exact projectors. For an input of weight $(0,2)$, one may use

$$\prod_{\ell=1}^{2}\left(1-\frac{F_L E_L}{\ell(\ell+1)}\right), \qquad \prod_{\ell=2}^{12}\left(1-\frac{F_R E_R}{\ell(\ell+1)-2}\right).$$

The raising operator acts first in every FE factor.

Certificate input 3.2 (Lower identities and the actual triplet)**.** The lower-closure artifact contains 3,970 completed exact identities. The actual P_4 vectors satisfy 18 current-zero-mode identities and six independently checked inclusions

$$L(1)P_4 \subset S_3, \qquad L(2)P_4 \subset S_2.$$

These direct Virasoro checks are available before the weight-four gate.

The two gates left for the all-weight argument are

$$(S_2)_0 S_3 \subset W_4 \qquad \text{(Gate 4)}, \tag{9}$$

$$(S_2)_0 P_4 \subset N_5 \qquad \text{(Gate 5)}. \tag{10}$$

4 Gate 4: exact seeds, raising, and singlet recovery

Put $P = S_3/N_3$, of independently known dimension 322. For a fixed joint-highest primitive field $a \in S_2$, the map

$$f_a : P \longrightarrow B_4/W_4, \qquad [p] \longmapsto [a_0 p]$$

is well defined because the lower relations give $a_0 N_3 \subset N_4$. Current covariance implies that its kernel is stable under both commuting raising operators E_L, E_R.

Lemma 4.1 (Raising saturation). *If $C_a \subset \ker f_a$ and $P = E_L P + E_R P + C_a$, then $f_a = 0$.*

Proof. On $M = P/\ker f_a$ the hypothesis says $M = E_L M + E_R M$. The commuting ideal (E_L, E_R) acts nilpotently because all current weights are bounded. Iterating gives $M = (E_L, E_R)^r M = 0$ for sufficiently large r. □

The exact raising coordinates comprise 644 images. Their rank is 294. For each of nine independent highest left fields, the completed seed sets and these images have rank 322. A nonzero minor at the good split prime $p = 1{,}000{,}003$ proves a nonzero characteristic-zero minor. This proves spanning because the exact ambient quotient dimension is already 322. It does not give an upper rank for an unknown weight-four normal space.

The original exact run supplies 2,145 seed identities. The saturation calculation asks for 130 further inputs; 118 have direct exact residual certificates. The remaining twelve outputs have weight $(0, 0)$. An exact replay of the known kernel and twelve additional identities at output weight $(0, 2)$ shows that both raising currents kill each of these twelve remaining classes modulo W_4.

Lemma 4.2 (Recovery from a joint-singlet projection). *Let $t \in B_4$ have weight $(0, 0)$, with $E_L t, E_R t \in W_4$. Define*

$$\Pi = \prod_{j \in \{2,4\}} \left(1 - \frac{4F_L E_L}{j(j+2)}\right) \prod_{j \in \{2,4,\ldots,24\}} \left(1 - \frac{4F_R E_R}{j(j+2)}\right). \tag{11}$$

If Πt is a linear combination of Πn_i, with $n_i \in N_4$, then $t \in W_4$.

Proof. Both W_4 and N_4 are stable under current zero modes. Every FE factor kills $[t] \in B_4/W_4$, so $[\Pi t] = [t]$. Each Πn_i remains in $N_4 \subset W_4$. Thus $[t] = 0$. Equivalently, the weight-zero highest class is a singlet in the finite semisimple current module. No dimension assertion about the selected normal columns is needed. □

The prime search selects 21 independent projected normal columns. All twelve projected identities have complete residual zero over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$. The final verifier repeats those residuals by independent integer denominator clearing, authenticates the raising premise, and repeats all nine saturation ranks. By Lemmas 4.1 and 4.2, all $9 \cdot 322 = 2898$ required highest-left identities hold. Current covariance gives the charged descendants. The second neutral field follows from $\omega_B - \omega_{K_L} - \omega_{K_R}$ minus the retained neutral field. This proves (9) with the certified lower controls.

5 Gate 5 from Jacobi identities

The proof uses the full normal spaces in (7)–(8), not only the selected solver columns. Before using Gate 5, the lower controls and Gate 4 give

$$(S_2)_0 N_4 \subset N_5, \qquad (N_3)_0 S_3 \subset N_5. \tag{12}$$

The first follows by zero-mode derivation and skew reordering. For the second, $(Ta)_0 = 0$ and, for a current x,

$$(x_{-1}a)_0 p = \sum_{i\geq 0} x_{-1-i}(a_i p) + \sum_{i\geq 0} a_{-1-i}(x_i p). \tag{13}$$

All inner products belong to the required lower spaces. To make the translation reduction explicit, repeatedly use

$$x_{-r}v = \frac{T(x_{-(r-1)}v) - x_{-(r-1)}Tv}{r-1}, \qquad r \geq 2.$$

The base terms are precisely those in (8); the same identity applies to $a_{-r}v$ with $a \in S_2$. Thus no unproved P_3-by-P_3 closure enters (12).

If $P_4 \subset N_4$, Gate 5 already follows. Otherwise $Q = W_4/N_4$ is the irreducible current triplet of highest weight $(0, 2)$, since the actual P_4 is irreducible. Let q_-, q_0, q_+ be its weight basis, and set

$$\beta(a, p) = [a_0 p] \in Q, \qquad F(a, q) = [a_0 \widetilde{q}] \in B_5/N_5.$$

The second map does not depend on the representative $\widetilde{q} \in W_4$, by (12).

Lemma 5.1 (Jacobi reduction)**.** *If $a, b \in S_2$ satisfy $a_0 b \in N_3$, then for $p \in S_3$,*

$$F(a, \beta(b, p)) = F(b, \beta(a, p)). \tag{14}$$

Proof. The difference is the class of $(a_0 b)_0 p$, which vanishes by (12) and $[a_0, b_0] = (a_0 b)_0$. □

5.1 The charged families

Let $a_0, \ldots, a_7$ denote the eight charged highest fields, with current weights

$$(0, 4), (0, 8), (0, 8), (0, 12), (1, 1), (1, 5), (1, 7), (1, 13).$$

In a product such as $(a_i)_0 p$, the parenthesized subscript denotes the field label and the next subscript denotes its mode. The saved exact relations give

$$\beta(a_3, p_3) = c_3 q_-, \qquad \beta(a_6, p_6) = c_6 q_-, \qquad c_3 c_6 \neq 0,$$

where $p_3 = \texttt{r3_000910}$ has weight $(0, -14)$, and $p_6 = \texttt{r3_000899}$ has weight $(-1, -9)$. The coefficients are authenticated in the final Jacobi certificate. Exact combinations of the lower relations establish $(a_i)_0 a_j \in N_3$ for the following pairs:

i	j	Input	Weight of $(a_i)_0 p_j$
0	3	p_3	$(0, -10)$
1,2	3	p_3	$(0, -6)$
3	6	p_6	$(-1, 3)$
4	6	p_6	$(0, -8)$
5	3	p_3	$(1, -9)$
6	3	p_3	$(1, -7)$
7	3	p_3	$(1, -1)$

Every listed output weight lies outside Q, so $\beta(a_i, p_j) = 0$. Equation (14) and $c_j \neq 0$ give $F(a_i, q_-) = 0$. Raise this resulting zero with a_i held highest to obtain the other two right inputs. Current covariance gives all left descendants.

Remark 5.2. The weight exclusion is used only at the displayed bottom inputs. It need not hold after raising p_j; using it there would be an incorrect stronger assertion. We raise the proved zero identity instead.

5.2 The neutral family

Let C be the full current span, inside $P = S_3/N_3$, of brackets of charged weight-two fields. The charged-span certificate reconstructs 28 actual charged seeds from the exact lower relations. Together with the 644 lowering images, of rank 294, these have rank 322. The independent verifier also checks both $\mathfrak{sl}_2$ commutators and all cross-axis commutators against the saved exact raising matrices. Since C is a current submodule, the nilpotent-ideal argument gives

$$P = F_L P + F_R P + C \quad \Longrightarrow \quad P/C = 0.$$

Once again the finite-prime full rank is a positive spanning witness inside an independently exact 322-dimensional space.

For the retained neutral singlet e, write

$$e_0 a_3 = \sum_i t_i (x_i)_0 y_i + n, \qquad n \in N_3,$$

with charged x_i, y_i. Apply zero modes to p_3. By Jacobi, the charged conclusion and (12),

$$(e_0 a_3)_0 p_3 = \sum_i t_i [(x_i)_0, (y_i)_0] p_3 + n_0 p_3 \in N_5.$$

Indeed Gate 4 puts each inner product in W_4, and the next charged product is controlled on both N_4 and P_4. A second Jacobi identity gives

$$e_0((a_3)_0 p_3) = (a_3)_0 (e_0 p_3) + (e_0 a_3)_0 p_3 \in N_5.$$

The nonzero coefficient of q_- proves the neutral bottom identity; covariance gives the triplet. The other neutral field is obtained from the total stress minus the two affine Sugawara stresses and e. This proves Gate 5. Its 24 independent highest channels comprise 23 charged channels and one neutral channel; they are consequences of the displayed identities, not 24 separately recomputed grade-five lattice products.

6 From the finite gates to properness

The order of deductions prevents a circular use of Virasoro closure. The direct controls in Certificate 3.2 first give $L(1)W_4 \subset S_3$ and $L(2)W_4 \subset S_2$. For quasiprimary $a \in S_2$, Gate 4 then gives, for $p \in P_3$,

$$\begin{aligned} 2a_1 p &= L(1)(a_0 p) - a_0 L(1) p \in S_3, \\ a_2 p &= L(1)(a_1 p) - a_1 L(1) p \in S_2, \\ 2a_3 p &= L(2)(a_1 p) - a_1 L(2) p \in K. \end{aligned}$$

The next mode is scalar. Affine normal left fields follow by iterate identities. The actual provenance of P_4 and

$$k_1(a_0 p) = (k_0 a)_1 p + (k_1 a)_0 p + a_0 (k_1 p)$$

give $K_1 P_4 \subset S_3$. Current zero-mode stability is already known; higher modes follow from the Virasoro commutators. Consequently $L(1)N_5 \subset W_4$ and $L(2)N_5 \subset S_3$. Gate 5 now yields, for $p \in P_4$,

$$\begin{aligned} 2a_1 p &= L(1)(a_0 p) - a_0 L(1) p \in W_4, \\ a_2 p &= L(1)(a_1 p) - a_1 L(1) p \in S_3, \\ 2a_3 p &= L(2)(a_1 p) - a_1 L(2) p \in S_2, \\ a_4 p &= L(2)(a_2 p) - a_2 L(2) p \in K. \end{aligned}$$

The next mode is scalar. The 30 earlier prime-only a_1P_4 tests are therefore derived exactly here; they are not an omitted gate.

All hypotheses of Theorem 3.1 hold for K, S_2, S_3, P_4. Their normal span is a vertex subalgebra of U containing K, u, ω_B, hence equals U. A normal word in positive-weight generators cannot acquire weight one unless it is a weight-one generator. Thus $U_1 = K$, proving Proposition 1.1.

REVIEW — F1: what the finite certificate establishes

This proof of properness has the strongest computational support in the manuscript. It still relies on correctness of the lattice mode engine, the saved actual-vector recipes, and the stated algebraic lemmas. Hashes authenticate the records; hashes alone do not prove those lemmas or independently reconstruct every earlier mode calculation. No claim of a proof-assistant check or a second implementation of the entire construction is intended. In particular, $\dim U_4 = 6010$ is not used as an input to this proof.

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7 Regularity from a positive conformal-vector identity

This section concerns the actual PCT-stable subalgebra $U \subset B$ defined in the construction. Its conclusions do not require properness. We use the convention $Y(a,z) = \sum_{n\in\mathbb{Z}} a_n z^{-n-1}$ and write $L^e(n) = e_{n+1}$ for the Virasoro operators of a conformal vector e. The input from the local construction is the identity

$$\omega_U = \omega_{A_{1,1}} + \frac{3403}{4563}\omega_{A_{1,39}} + \frac{25}{6}\overline{p} + \frac{25}{26}\overline{z}. \tag{15}$$

Here p and z are the conformal vectors of actual embedded copies of the three-state Potts algebra $K(\mathfrak{sl}_2, 3)$ and the parafermion algebra $K(\mathfrak{sl}_2, 13)$; a bar denotes normalized averaging over the compact current group. Each subalgebra is regular and unitary, and its intrinsic grading agrees with the ambient grading on its states. The averages take place in the finite-dimensional space U_2. Finite-dimensional convexity replaces each average by a finite convex combination of conjugate conformal vectors, so that

$$\omega_U = \sum_{i=1}^{N} c_i e_i, \qquad c_i > 0. \tag{16}$$

Zero coefficients, if any, are omitted. All simple modules of each regular subalgebra A_i have nonnegative lowest weights, with zero occurring only for its vacuum module.

REVIEW — R1: the embedded regular factors are local inputs

The existence and normalization of the Potts and parafermion fields, their inclusion in the generated U, and (15) are imported from the earlier construction and the proof records `hi_cft_20260923_next_structure_c2_energy_bound.md` and `hi_cft_20260923_next_structure_top_bound.md`. They are not consequences of the finite closure theorem alone. This manuscript gives the deductions from those inputs, but does not reproduce their full field-identification calculation. A review of regularity must verify these embeddings, not only the displayed positive coefficients.

Lemma 7.1 (Orthogonal energy bound)**.** *Let V be a unitary VOA of CFT type admitting an identity of the form (16), with the properties just stated. Then V is C_2-cofinite.*

Proof. For an ordinary simple A_i-module M, put

$$C_{2,A_i}(M) = \operatorname{span}\{a_{-2}m : a \in A_i,\ m \in M\}.$$

These quotients are finite-dimensional by Abe–Buhl–Dong, Proposition 5.2. There are finitely many simple A_i-modules. Hence there is a finite number D_i bounding the intrinsic energy in all their C_{2,A_i} quotients; this includes their lowest weights. Regularity decomposes the weak A_i-module V into ordinary simples, and the unitary form permits an orthogonal decomposition. Since

$$C_{2,A_i}(V) \subset C_2(V),$$

the self-adjoint intrinsic energy satisfies

$$\langle L^{e_i}(0)v, v\rangle \le D_i\|v\|^2 \qquad (v \in C_2(V)^\perp).$$

For $v \in V_n \cap C_2(V)^\perp$, the conformal-vector identity gives

$$n\|v\|^2 = \sum_i c_i\langle L^{e_i}(0)v, v\rangle \le \Big(\sum_i c_i D_i\Big)\|v\|^2.$$

Thus the graded quotient $V/C_2(V)$ vanishes above a finite weight. No simultaneous diagonalization of the different energy operators is required. The same proof applies directly to averages, since every conjugate factor has the same bound D_i. □

Lemma 7.2 (Lowest weights and projectivity of the vacuum)**.** *Every simple ordinary U-module has nonnegative lowest conformal weight, and the only simple module of lowest weight zero is U. Moreover U is projective in the category of grading-restricted generalized modules.*

Proof. Let E be the lowest space of a simple ordinary module of weight h. Every mode of an A_i state that lowers intrinsic weight also lowers ambient weight, and therefore annihilates E. Thus $E \subset \Omega_{A_i}(M)$, the Zhu highest space. Restriction initially gives a weak A_i-module; regularity decomposes it into ordinary simple modules. The highest space of an ordinary simple is its top, so each $L^{e_i}(0)|_E$ is diagonalizable with nonnegative real spectrum. Taking traces, rather than assuming the operators commute, gives

$$h\dim E = \sum_i c_i \operatorname{Tr}_E L^{e_i}(0) \ge 0. \tag{17}$$

If $h = 0$, every summand vanishes. Hence every partial energy is zero on E, which lies in sums of vacuum tops for every A_i. Consequently $L^{e_i}(-1)E = 0$, and (16) gives $L_U(-1)E = 0$. Any nonzero $v \in E$ is vacuum-like: translation covariance and lower truncation show that $Y_M(a,z)v$ has no negative powers of z. The map $a \mapsto a_{-1}v$ is a nonzero module map $U \to M$, hence an isomorphism.

Now let $\pi : E \twoheadrightarrow U$ be an epimorphism in the generalized module category. C_2-cofiniteness gives finite length, and the preceding argument excludes negative weights in all its composition factors. The finite generalized weight-zero space $E_{[0]}$ therefore has no lower-weight target for a lowering mode. It is again in every $\Omega_{A_i}(E)$. On this space $L_U(0)$ is nilpotent, so

$$0 = \operatorname{Tr}_{E_{[0]}} L_U(0) = \sum_i c_i \operatorname{Tr}_{E_{[0]}} L^{e_i}(0).$$

Diagonalizability and nonnegativity of each partial energy imply it is zero. The vacuum-top argument gives $L_U(-1)E_{[0]} = 0$. Choose $v \in E_{[0]}$ with $\pi(v) = \mathbf{1}$. Then

$$s : U \longrightarrow E, \qquad s(a) = a_{-1}v, \qquad \pi s = \mathrm{id}_U,$$

is a section. Every epimorphism onto the vacuum splits. □

Proposition 7.3 (Regularity of the actual generated algebra)**.** *The VOA U is strongly rational.*

Proof. Simplicity, CFT type, and self-contragredience follow from the unitary construction. The preceding lemmas supply C_2-cofiniteness and a projective vacuum. The category $\mathcal{C}$ of grading-restricted generalized modules is finite abelian and braided, with right-exact tensor product and Grothendieck–Verdier duality whose dualizing object is $U' = U$; the relevant VOA tensor-category hypotheses are reviewed in McRae, Sections 2.3 and 3.

Here is the remaining categorical argument. Realize $\mathcal{C}$ as finite-dimensional modules for a finite-dimensional algebra, denote underlying vector-space dimension by d_0, and let P correspond to its regular module. Every Z is a quotient of $P^{\oplus d_0(Z)}$. For fixed X, right exactness gives

$$d_0(Z \otimes X) \le C_X d_0(Z), \qquad \dim \mathrm{End}(X^{\otimes n}) \le d_0(\mathbf{1})^2 C_X^{2n} < n!$$

for large n, where $C_X = \max\{1, d_0(P \otimes X)\}$. For nonzero indecomposable X, Grothendieck–Verdier evaluation $DX \otimes X \to \mathbf{1}$ is nonzero, hence surjective onto the simple unit, and splits by projectivity. Thus X is non-negligible and has moderate growth in the precise sense of Etingof–Penneys, Theorem 1.1; that theorem proves X rigid. Finite direct sums give rigidity of all objects. Tensoring is now exact, and

$$\mathrm{Hom}(X, -) \cong \mathrm{Hom}(\mathbf{1}, - \otimes X^*)$$

is exact. Every object is projective, so $\mathcal{C}$ is semisimple and U is rational. This does not assume complete unitarity of its modules. The local argument is recorded in `hi_cft_20260923_next_structure_rationality.md`. □

8 Branching and the boundary fusion ring

From this point we use the properness conclusion and $\dim U_1 = 6$ of the finite closure argument. Before that conclusion is supplied, every statement in this section is conditional on the proper branch.

8.1 A universal top bound and actual lattice branching

The sharper bound uses an actual regular subalgebra $A_L \subset U$. In the notation of the local construction, a rational lattice–Potts core $\mathcal{T}$ and $K(\mathfrak{sl}_2, 13)$ have pointed charges $(j, k) \in (\mathbb{Z}/13)^2$ with quadratic form $-(j^2 + k^2)/13$. Exact weight-two vectors witness the simple-current charge $(5, 1)$ (up to reversing the second charge). Locality and multiplicity one in a connected extension force its entire isotropic line, yielding

$$A_L = \bigoplus_{k=-5j} \mathcal{T}_{1,j} \otimes \mathrm{PF}_{0,k} \subset U, \qquad \mathrm{Rep}(A_L) \simeq \mathrm{Fib} \boxtimes \mathrm{SO}(3)_{13}.$$

The field witness and extension argument are in `hi_cft_20260923_next_structure_order13_core.md` and `hi_cft_20260923_next_structure_core_extension.json`. There are fourteen simple tops, labelled $x_a = (1, a)$ and $y_a = (\tau, a)$ for $0 \le a \le 6$.

Let E be any simple U-module top and write x_a, y_a also for the nonnegative integer multiplicities of these A_L tops in E. Set $D = \dim E$. For each top, let L be the trace of the left affine energy, $Q = \mathrm{Tr}\, H_R(0)^2$, and P, Z the Potts and parafermion energy traces. Its contribution to (17) is

$$W = L + \frac{83}{6084} Q + \frac{25}{6} P + \frac{25}{26} Z. \tag{18}$$

This is a trace formula on a complete right $\mathfrak{sl}_2$ representation, not an assertion about a single partial-energy eigenvector. The exact dimensions and trace contributions are

top	d	$4680W$	top	d	$4680W$
x_0	1	0	y_0	1	7800
x_1	1	600	y_1	7	36960
x_2	5	10440	y_2	15	110880
x_3	9	56640	y_3	2	4080
x_4	2	5520	y_4	14	81480
x_5	14	148560	y_5	6	18240
x_6	6	54240	y_6	2	4920

Complete right $\mathfrak{sl}_2$ characters imply $y_2 \geq x_5$ and $y_4 \geq x_5 + x_6$: these compare charges $10, 12$ and charges $7, 9$, respectively, with the common left doublet removed in the second comparison. Direct comparison of the displayed table gives

$$\begin{aligned} 4680(\tfrac{17}{10}D - W_{\text{total}}) = {} & 8460(y_2 - x_5) + 28716(y_4 - x_5 - x_6) \\ & + 7956x_0 + 7356x_1 + 29340x_2 + 14964x_3 + 10392x_4 \\ & + 22212x_6 + 156y_0 + 18732y_1 + 11832y_3 \\ & + 1188y_4 + 29496y_5 + 10992y_6 \geq 0. \end{aligned} \tag{19}$$

Since $W_{\text{total}} = hD$, this proves $0 \leq h \leq 17/10 < 2$ for *every* simple U-module, without assuming it is unitarizable.

REVIEW — R2: the core-top table and finite-top actions

The inequality (19) can be checked directly from the displayed table. Deriving that table, including all minimizing charge representatives, uses `hi_cft_20260923_next_structure_order13_top_bound.md`. The construction of $\mathcal{T}$, the thirteen-charge extension, and completeness of the resulting fourteen-module list are substantial local inputs whose full derivations are not repeated here. Likewise, the finite-top irreducibility used below is imported from exact field actions, particularly `hi_cft_20260923_next_81top_all.json`; it should not be replaced by numerical cyclicity alone.

The strongly rational restriction $B\downarrow_U$ is semisimple. Its simple summands have integer lowest weights, hence only weights $0, 1$ are possible. The vacuum occurs once, while the complementary current space has dimension $86 - 6 = 80$. Its certified irreducible zero-mode action therefore implies

$$B\downarrow_U = U \oplus X, \qquad h_X = 1, \qquad \dim X_{\text{top}} = 80. \tag{20}$$

The four mixed nontrivial $E_6 \oplus A_2$ lattice sectors start at weight 1, with 81-dimensional tops. Their next possible top weight would be 2, excluded by (19). Their exact top-action certificates use a one-dimensional Cartan eigenline, positive-metric adjoint identities, and a cyclic span of dimension 81. The adjoint identities make the generated operator algebra a star algebra; orthogonal complements are therefore invariant, and the unique eigenline plus cyclicity proves irreducibility.

Write the actual lattice extension as $H = B \oplus J \oplus J^2$. Its restriction is consequently

$$H\downarrow_U = \mathbf{1} \oplus X \oplus Y \oplus Y^*, \qquad h_Y = 1, \quad \dim Y_{\text{top}} = 81. \tag{21}$$

At this stage $Y \cong Y^*$ has not been assumed. The three types $\mathbf{1}, X, Y$ are distinguished by their lowest spaces.

8.2 The rank-four or rank-six alternative

Put $\mathcal{C} = \mathrm{Rep}(U)$ and $\mathcal{F} = \mathcal{C}_H$, the category of *all* right modules over the extension algebra H. The strict positivity of nonvacuum lowest weights gives the positive spherical structure through character asymptotics. Since H is holomorphic, it is Lagrangian and the DMNO center correspondence gives $\mathcal{C} \simeq Z(\mathcal{F})$. Write $s = \mathrm{FPdim}_{\mathcal{C}} B$. Extension dimension identities give

$$d_X = s - 1, \qquad d_Y = d_{Y^*} = s, \qquad \mathrm{Dim}\,\mathcal{C} = 9s^2, \quad \mathrm{Dim}\,\mathcal{F} = 3s.$$

Pseudo-unitarity of $\mathcal{F}$ follows from that of its center and the squared global-dimension and Frobenius–Perron-dimension identities. No C^* structure is needed in this section.

For center induction I and forgetful functor F, one has $H = I(\mathbf{1})$ and $FI(\mathbf{1}) = \bigoplus_{a \in \mathrm{Irr}\,\mathcal{F}} a \otimes a^*$. Adjunction therefore gives

$$\mathrm{rank}\,\mathcal{F} = \dim \mathrm{End}_{\mathcal{C}}(H) = \begin{cases} 4, & Y \ncong Y^*, \\ 6, & Y \cong Y^*. \end{cases} \tag{22}$$

The actual order-three automorphisms of H/B inject into $\mathrm{Inv}(\mathcal{F})$. This injection is already spherical: under the composition/convolution description of $\mathrm{End}(I(\mathbf{1}))$ in Bischoff–Jones, Lemma 3.1 and Corollary 3.4, an algebra automorphism and its inverse are convolution idempotents. They are positive sums of the primitive idempotents $d_a[a]$; their fusion product is $[\mathbf{1}]$. Positivity forces singleton supports consisting of inverse invertibles.

The earlier pointwise automorphism calculation gives $\mathrm{Aut}(B/U) = 1$. If rank six had six invertibles, then $3s = 6$, so X would be invertible and $B = \mathbf{1} \oplus X$ would have a nontrivial sign automorphism, a contradiction. Thus there are exactly three invertibles in either rank. Their associator is also determined: the local B-module category has metric group

$$(\mathbb{Z}/3)^2, \qquad q(a,b) = \frac{2a^2 + b^2}{3} \pmod 1.$$

The actual extension H/B selects an isotropic line. Forming its modules gives an embedded untwisted $\mathrm{Vec}_{\mathbb{Z}/3}$ subcategory of $\mathcal{F}$. Merely knowing the group of invertibles would not determine this associator.

8.3 Excluding rank four: scope of the longer local argument

At rank four the fusion ring is near-group:

$$a^3 = 1, \qquad a\rho = \rho a = \rho, \qquad \rho^2 = 1 + a + a^2 + m\rho.$$

The published Dong–Chen–Wang, Theorem 6.2, restricts m to $0, 2, 3, 6$. For $m = 0$, $s = 2$ and the sign-automorphism contradiction just given applies. For $m = 2$, the categorical reduction of Etingof–Gelaki–Ostrik, Theorem 7.1 and Corollary 7.4 supplies the twisted A_4 double. Its electric regular algebra gives a holomorphic $c = 8$ extension H' with $U = (H')^{A_4}$, and Dong–Mason, Theorem 1, identifies H' with V_{E_8}. The order-two and order-three adjoint traces in E_8 are bounded below by -8 and -4, respectively, as follows from the affine Dynkin-diagram classification. Therefore

$$\dim(E_8^{A_4}) \geq \frac{248 + 3(-8) + 8(-4)}{12} = 16,$$

contradicting $\dim U_1 = 6$.

For clarity, the remaining exclusions use universal spherical induction arguments, not a classification limited to unitary examples. For $m = 3k$, induction yields integers $p, q \geq 0$

with $p+q=k$. At $k=1$ the pointed factor and two indicator moments force the modular representation of $G_{2,3}\otimes A_{2,1}$ or its reversal. Harmonic theta-series pole cancellation in these two orientations gives

$$\dim U_1 = 15+A+10B \quad \text{or} \quad 14+42A, \qquad A,B\in\mathbb{Z}_{\geq 0}.$$

At $k=2$, the balanced pattern $p=q=1$ forces, by signed Galois symmetry, a row of the center's S matrix whose Verlinde coefficient is

$$N_{XX}^{X} = \frac{97}{72}+\frac{32}{72}-\frac{33}{72} = \frac{4}{3}.$$

The unbalanced pattern $\{p,q\}=\{0,2\}$ admits a weighted character projection. Its three-dimensional block is

$$S_v = -\frac{1}{2}\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & -2 & 1 \end{pmatrix}, \qquad T_v = \operatorname{diag}(1,-i,-1)$$

before the common central-charge factor; reversal conjugates T_v. Harmonic theta-series subtraction gives the two alternatives

$$\dim U_1 = 8+A \quad \text{or} \quad 5+12P, \qquad A,P\in\mathbb{Z}_{\geq 0}.$$

None of these formulas allows six currents. The parameters are actual weighted polar top dimensions, so their integrality and positivity do not require linear independence of all simple characters.

REVIEW — R3: compressed rank-four exclusions

The preceding paragraph is a summary, not a full proof of the universal $m=3,6$ exclusions. The derivation of the complete modular representation for $m=3$ is in `hi_cft_20260923_next_m3_universal_data.md`, with an independent audit in `hi_cft_20260923_next_m3_universal_independent_audit.md`. The $m=6$ arguments are in `hi_cft_20260923_next_structure_m6_balanced_exclusion.md` and `hi_cft_20260923_next_structure_m6_universal_projection.md`. The coefficient calculations and pole-cancellation arguments are in `hi_cft_20260923_next_structure_m3_character_obstruction.md` and `hi_cft_20260923_next_structure_m6_character_obstruction.md`. These are new local deductions, not statements of the cited rank-four classification theorem. Their universal induction reductions and theta normalizations remain an explicit review dependency of this condensed manuscript. In particular, the older character notes alone only exclude specified model representations; the later universal reductions are essential. The pointwise claim $\operatorname{Aut}(B/U)=1$ and the $m=2$ electric-extension argument also rely on the longer record `hi_cft_20260923_next_structure_rank46.md` and its replacement spherical audit `hi_cft_20260923_next_structure_final_dependency_audit.md`.

8.4 The rank-six ring and exclusion of higher multiplicity

Rank four being excluded, (22) gives $H=\mathbf{1}\oplus X\oplus 2Y$. The fusion algebra is

$$K_0(\mathcal{F})\otimes\mathbb{C}\cong\mathbb{C}\oplus\mathbb{C}\oplus M_2(\mathbb{C}),$$

by the composition algebra of $\operatorname{End}_{\mathcal{C}}(H)$. Hence it is noncommutative. The three invertibles act transitively on the three noninvertibles: otherwise both nontrivial idempotents of $\mathbb{C}[\mathbb{Z}/3]$ split off, leaving a four-dimensional ideal with a dimension character, incompatible with the M_2 block. The left and right cyclic actions must be inverse; equality would make the fusion

ring commutative. Duality then fixes every noninvertible, and reciprocity and associativity give

$$a^3 = 1, \qquad \rho a = a^2\rho, \qquad \rho^2 = 1 + m(\rho + a\rho + a^2\rho), \qquad m \geq 1. \tag{23}$$

Here are the essential numerical identities in the local multiplicity exclusion. Put

$$d = \frac{3m + \sqrt{9m^2+4}}{2}, \quad s = 1 + d^2, \quad \Delta = 9m^2 + 4, \quad R = \rho + a\rho + a^2\rho.$$

Center induction, with the untwisted pointed subcategory, yields

$$I(1) = 1 + X + 2Y, \quad d_X = d^2, \quad d_Y = s, \qquad I(a) = I(a^2) = C_0 + C_1 + C_2,$$

where $\theta_X = \theta_Y = 1$, $d_{C_j} = s$, and the C_j twists are $1, \zeta_3, \zeta_3^2$. All remaining central simples Z_j have $F(Z_j) = n_j R$, $n_j \in \mathbb{Z}_{>0}$. Global dimension and the first two Ng–Schauenburg indicators give

$$\sum_j n_j^2 = 4m^2 + 2,$$

$$\sum_j n_j^2 \theta_j = -\frac{m^2 + m\sqrt{\Delta}}{2}, \tag{24}$$

$$\sum_j n_j^2 \theta_j^2 = \frac{-m^2 + (2\varepsilon - m)\sqrt{\Delta}}{2}, \qquad \varepsilon = \nu_2(\rho) \in \{1, -1\}. \tag{25}$$

These follow from the center-twist trace formula in Ng–Schauenburg, Theorem 4.1. They concern a multiset of exactly $4m^2 + 2$ roots, repeating θ_j precisely n_j^2 times.

Write $\Delta = t^2 D$ with D squarefree. One has $D \equiv 1 \pmod 3$ and $7 \nmid D$. Except when $D = 10$, the elementary prime-factor estimate $\varphi(2D)/\sqrt{D} \geq 12/\sqrt{13}$ and Larson's root-count theorem applied to (24) give

$$4m^2 + 2 \geq \frac{mt}{2}\varphi(2D) \geq \frac{6m\sqrt{9m^2+4}}{\sqrt{13}}.$$

For $m > 1$ this is impossible, since

$$36m^2(9m^2 + 4) - 13(4m^2 + 2)^2 = 4(m^2 - 1)(29m^2 + 13) > 0.$$

For $D = 10$, a root multiset with sum $-a - b\sqrt{10}$, with $a, b \in \mathbb{Z}_{\geq 0}$, has at least $a + 8b$ members. In the equality case its squared-root sum is a, provided that sum lies in $\mathbb{Q}(\sqrt{10})$. To see the equality condition, average over the Galois group fixing $\sqrt{10}$. Rational contributions cost at least one root per negative unit, with equality only for -1; quadratic contributions cost at least eight roots per unit, with equality only for a primitive-40 orbit. That orbit has squared sum zero. The hypothesis on the second sum permits undoing the average.

In our application m, t are even and $a = m^2/2$, $b = mt/2$. The bound

$$4m^2 + 2 \geq \frac{m^2}{2} + 4mt$$

excludes even $m \geq 4$. The remaining $m = t = 2$ is the equality case, so the squared sum must be 2, whereas (25) gives -2 or $-2 - 4\sqrt{10}$. Thus $m = 1$. The full conductor-orbit proof, including imprimitive orbits and the second-moment hypothesis, is retained in `hi_cft_20260923_next_structure_rank6_indicators.md` and `hi_cft_20260923_next_rank6_root_audit.md`.

Consequently

$$d = \frac{3+\sqrt{13}}{2}, \qquad s = \frac{13+3\sqrt{13}}{2}, \qquad d_X = \frac{11+3\sqrt{13}}{2}. \tag{26}$$

For $m = 1$, (24) has six roots with sum $(-1-\sqrt{13})/2$. The quadratic-period equality case forces the six distinct nonresidue thirteenth roots, so $n_j = 1$ and the center has twelve simples. Its dimensions and twist exponents are

number	1	1	4	6
dimension	1	d^2	s	$3d$

$$0, 0, 0, 0, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{2}{13}, \tfrac{5}{13}, \tfrac{6}{13}, \tfrac{7}{13}, \tfrac{8}{13}, \tfrac{11}{13}.$$

REVIEW — R4: conductor-orbit bounds and equality cases

The reduction to $m = 1$ and the count of twelve central simples use more than the displayed indicator identities. The prime-factor estimate, the exceptional $\mathbb{Q}(\sqrt{10})$ equality case, and the quadratic-period equality case at $m = 1$ are compressed here. Their full conductor-orbit arguments, including imprimitive roots and multiplicities, are in `hi_cft_20260923_next_structure_rank6_indicators.md`, `hi_cft_20260923_next_rank6_root_audit.md`, and `hi_cft_20260923_next_structure_rank6_center.md`. These local deductions need review independently of the cited indicator formula. The later net-sector count inherits this dependency.

The complete S, T data follow from Gannon–Schopieray–Yadav, Theorem A(2), equivalently Theorem 5.1: its relevant statement assumes a complex spherical untwisted HI category. The auxiliary metric group has prime order 13 and its negative quadratic Gauss sum fixes its isometry class. This identifies the standard Haagerup modular data. It does not by itself identify a braided category; that stronger conclusion is treated separately after complete unitarity.

9 The complete character vector

Let $q = e^{2\pi i\tau}$. Character exponents are $h - c/24 = h - 1/3$. Using the preceding twist list and strict positivity of nonvacuum lowest weights, the only allowed nonvacuum pole is $\gamma q^{-7/39}$ in the twist-$2/13$ coordinate. Here $\gamma \in \mathbb{Z}_{\geq 0}$ is its top dimension if that module starts at $2/13$, and is zero if its lowest weight is shifted upward. The vacuum pole is $q^{-1/3}$ with coefficient one.

Lemma 9.1 (Uniqueness from the principal part)**.** *Two character vectors for this modular representation with the same γ coincide in every weight.*

Proof. Their difference is holomorphic at infinity. The modular representation has finite image, and covariance transports this property to every cusp of a finite-index kernel. Each component is therefore a holomorphic function on a compact modular curve, hence constant. A constant vector must be fixed by both S and the *character* T matrix, which includes $e^{-2\pi i/3}$. The only T-fixed coordinate is the object of categorical twist $e^{2\pi i/3}$. Its S image has a nonzero vacuum component, so it cannot span an invariant line. The constant is zero. □

Evans–Gannon, Sections 1.4 and 2.4, equation (2.7), provide a fundamental matrix Ξ whose vacuum and d_3 columns have precisely the two principal parts above. Thus, in their labeling,

$$\boldsymbol{\chi} = \Xi_{:,0} + \gamma \Xi_{:,d_3}, \qquad \dim U_1 = 6 + 13\gamma.$$

The actual six-current result forces $\gamma = 0$. Their recursive construction of Ξ and the uniqueness lemma give the entire vector, not just a finite coefficient match. In particular,

$$\operatorname{ch} U = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{27}$$

The lowest weights and top dimensions are

label	h	$\dim M_{\mathrm{top}}$	label	h	$\dim M_{\mathrm{top}}$
0	0	1	d_1	6/13	7
b	1	80	d_2	11/13	42
a	1	81	d_3	15/13	119
c_0	1	81	d_4	5/13	5
c_1	1/3	3	d_5	7/13	13
c_2	2/3	27	d_6	8/13	14

In particular the final maximum is 15/13, improving the preliminary 17/10 bound. The proof record is `hi_cft_20260923_next_structure_conditional_characters.md`. Although this vector fixes every unflavoured homogeneous dimension, it does not determine the separate branching functions for the fourteen-module core A_L, and it was not used to prove properness.

10 From vacuum unitarity to complete unitarity

The exact strong-generation certificates concern the actual subVOA $U \subset B \subset H = V_{E_8}$. In particular, they establish $\dim U_1 = 6$ and $U \neq B$; they do not establish positivity on all modules or on fusion products. We now separate those questions. *Vacuum unitarity* is the positive invariant form inherited by the PCT-stable subVOA U. *Strong unitarity* means that all simple ordinary U-modules admit compatible positive invariant forms. *Complete unitarity* additionally includes positivity of the canonical Hermitian fusion forms, giving the VOA representation category its unitary modular tensor structure. An abstract ribbon equivalence to a unitary category does not, by itself, prove this last positivity statement.

We use the preceding algebraic analysis as input: regularity of U, the actual restrictions

$$B{\downarrow_U} = U \oplus X, \qquad H{\downarrow_U} = U \oplus X \oplus Y^{\oplus 2}, \qquad h_X = h_Y = 1, \tag{28}$$

the twelve simple algebraic U-modules, and the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring of the all-module category for the original extension. Every use of these inputs inherits the qualifications attached to their preceding proofs. Agreement of modular data alone is never used as a classification of tensor categories.

HIGH PRIORITY REVIEW — Analytic proof boundary

The local applications labelled A1–A3 below are **claims pending expert review**. Their proposed arguments are included so that the domains, geometric identifications, and counting steps can be checked. The cited external theorems are used with their stated hypotheses; the issue under review is whether our construction satisfies and matches those hypotheses. Thus complete unitarity, the unitary ribbon identification, and the specifically $\mathcal{H}_3$ extension remain conditional in this manuscript. The exact finite certificates are separate accepted inputs and are not weakened by this distinction.

10.1 The subnet and its sector count

Write $\mathcal{N}_U \subset \mathcal{N}_H$ for the actual conformal-net inclusion. The lattice VOA supplies strong locality and bounded localized vertex operators; the unitary conformal subalgebra inherits the

relevant local construction. Once regularity is available, Tener, Corollary 4.24, applies to the conformal inclusion and gives finite index. The holomorphic lattice net is completely rational; Longo, Theorem 24, gives complete rationality of the subnet. This finite-index argument does not require every U-module to be unitary.

HIGH PRIORITY REVIEW — A2: actual net branching and twelve sectors

Claim 10.1 (Net-sector count; pending expert review)**.** *The restriction of the holomorphic H net has precisely the three inequivalent irreducible $\mathcal{N}_U$ summands corresponding to (28), with multiplicities $1, 1, 2$. Its boundary fusion category $\mathcal{F}_{\mathrm{net}}$ has rank six, untwisted invertible subcategory $\mathrm{Vec}_{\mathbb{Z}/3}$, and the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring. Consequently*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}}), \qquad \#\,\mathrm{Irr}\,\mathrm{Rep}(\mathcal{N}_U) = 12.$$

The passage from algebraic summands to irreducible net summands, and the transfer of the actual automorphism and endomorphism information, must be verified at the level of closed local operators. A shared branching matrix does not identify two tensor categories.

Here is the proposed reduction of this claim. On finite-energy vectors of H, reconstructed U-fields are its original restricted fields. Orthogonal projections onto its actual simple summands should commute with their closures. The resulting Lagrangian algebra has endomorphism algebra

$$\mathrm{End}(A_{\mathrm{net}}) \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{29}$$

The center-induction description then gives rank six and a noncommutative complex fusion algebra for $\mathcal{F}_{\mathrm{net}}$. The actual order-three automorphisms of H/B supply three invertibles. If all six simples were invertible, the intermediate index would be $[\mathcal{N}_B : \mathcal{N}_U] = 2$. The connected intermediate algebra $1 + X$ would have a sign automorphism. Reconstruction on finite energies would turn it into an automorphism of B fixing U, contrary to the actual calculation $\mathrm{Aut}(B/U) = 1$. Thus the invertible group is exactly $\mathbb{Z}/3$. The pointed nine-sector B net and the actual H/B extension supply its untwisted associator class.

With these positive spherical inputs, the ring and indicator arguments apply independently on the net side. In particular, for $d = (3 + \sqrt{13})/2$ and $s = 1 + d^2$, the center calculation gives dimensions

$$1,\ d^2,\ \underbrace{s, s, s, s}_{4},\ \underbrace{3d, 3d, 3d, 3d, 3d, 3d}_{6}. \tag{30}$$

The last six twists have exponents $\{2, 5, 6, 7, 8, 11\}/13$. This is the proposed source of the net count twelve. It is not obtained by assigning the VOA character transformation matrix to the net category. The underlying deductions are recorded in `hi_cft_20260923_next_structure_rank46.md`, `hi_cft_20260923_next_structure_rank6_indicators.md`, and `hi_cft_20260923_next_structure_rank6_center.md`.

10.2 Reverse reconstruction and exhaustion

Henriques–Tener, Theorem 9.5 and Corollary 9.7, reconstruct unitary modules and identify their closed smeared fields. Remark 9.8 extends the construction beyond the finite-multiplicity hypothesis. This extension matters: finite-dimensional energy eigenspaces must not be silently assumed for the net representation one is trying to reconstruct.

HIGH PRIORITY REVIEW — A3: ordinaryness, irreducibility, and linear fullness

Claim 10.2 (Reverse module comparison; pending expert review)**.** *For this regular U, reconstruction sends each irreducible $\mathcal{N}_U$ sector to an irreducible ordinary unitary U-module. It preserves L_0 and induces a fully faithful linear $*$-functor on finite direct sums,*

$$\mathcal{R} : \mathrm{Rep}(\mathcal{N}_U) \longrightarrow \mathrm{Rep}^u(U).$$

No tensorator or compatibility with braiding is asserted here.

The projection argument below must establish common-core invariance and commutation with closed local fields, including their adjoints. These are our analytic deductions from reconstruction, not a quoted braided-equivalence theorem.

Let $\mathcal{K}$ be an irreducible net representation and let

$$M = \bigcup_{R>0} 1_{[0,R]}(L_0)\mathcal{K}$$

be the dense bounded-energy space appearing in the extended construction. Regularity decomposes this admissible weak U-module algebraically into ordinary simple modules. Choose a nonzero summand S, inherit its positive form, and let P be the Hilbert projection onto $\overline{S} \subset \mathcal{K}$.

The intended argument first uses the agreement of the two Virasoro actions, as in Henriques–Tener Corollary 9.6. Each energy space of S is finite-dimensional, so its closure adds no vector at a fixed energy. It follows that P preserves the bounded-energy domain and maps it into S. The unitary adjoint identities, tested against S, then give commutation with all modes on the common domain. The required closure argument is that P and $1 - P$ reduce the closed smeared fields and therefore the local algebras they generate. Irreducibility of the net sector implies $P = 1$, and the finite-dimensional energy spaces then give $M = S$. This proves ordinaryness and simplicity if the stated domain assertions hold.

For fullness, an algebraic isomorphism between simple unitary modules is unitary after multiplication by a scalar, by uniqueness of the positive invariant form. Its bounded extension should intertwine the closed fields by the same argument. Conversely, net intertwiners commute with the reconstructed modes. Semisimplicity handles finite direct sums. The detailed local source is `hi_cft_20260923_next_m3_unitarity_scope.md`.

Proposition 10.3 (Conditional strong unitarity)**.** *Assume Claims 10.1 and 10.2, and the preceding algebraic count of twelve simple U-modules. Then every simple U-module is unitarizable, and its character is the character of a net sector.*

Proof. Full faithfulness makes the twelve net simples pairwise inequivalent simple ordinary U-modules. There are exactly twelve algebraic simple modules, so this injection is a bijection. Each image carries its reconstructed positive form, and L_0 is preserved. Finite direct sums give the corresponding statement for ordinary modules. □

This counting step supplies essential surjectivity only as a linear comparison. Neither fusion compatibility, equality of categorical S matrices, nor positivity of VOA fusion forms follows from it.

10.3 The annulus compatibility claim

The next step concerns the specific geometric annulus system of Tener, Definition 3.18 and Corollaries 3.20–3.22. Let (B, A) be a localized pair for an interval I, with incoming factor A and outgoing factor B. Subscripts distinguish their actions on the vacuum and on a reconstructed module. For an insertion point $z \in \mathrm{int}(B, A)$, put

$$X_v(B, z, A) = \overline{B_U Y_U(v, z) A_U} \in \mathcal{N}_U(I).$$

The bar denotes the bounded extension when it exists. The compatibility required by Tener's definition of a module net representation is the specific operator identity in the following claim.

HIGH PRIORITY REVIEW — A1: new annulus compatibility lemma

Claim 10.4 (Annulus compatibility; pending expert review)**.** *For the actual geometric system above, choose matching paths and central lifts in the two annulus constructions. If M is an ordinary unitary module reconstructed from a sector π_M, then*

$$\pi_M\big(X_v(B, z, A)\big) = \overline{B_M Y_M(v, z) A_M}, \qquad v \in U, \quad z \in \mathrm{int}(B, A). \tag{31}$$

The right side exists as a bounded extension. The same matching in the vacuum identifies the net's geometric insertion with $X_v(B, z, A)$.

The points requiring expert verification are modulewise matching of the annulus systems and central lifts, density and core properties of the inverse-finite-energy domains, and holomorphic continuation on every component across a possible circle seam. Agreement of vacuum operators alone would not prove this claim.

We give the proposed proof in three steps. First, geometric localization must precede operator comparison. The thin-boundary parametrizations in the selected system agree outside I, and its common localizer is geometric. Its generating operators are univalent-semigroup factors $\exp(-tL(\rho))$ and positive-energy diffeomorphism or Möbius operators. The time-ordered integration in Henriques–Tener, formula (5.2), gives the proposed comparison with the geometric annulus action. Constant generators give the same exponential; thin factors must give the same diffeomorphism action. A common path and central lift are essential. Using the same localizing lift for both members of (B, A) is intended to cancel deck-transformation factors. This matching is required on every positive-energy module, not merely on the vacuum.

Second, let M^{fin} be the finite-energy vectors and define

$$\mathcal{D}_A = \{\xi \in \mathcal{K}_M : A_M \xi \in M^{\mathrm{fin}}\}, \qquad \mathcal{D}_{B^*} = \{\eta \in \mathcal{K}_M : B_M^* \eta \in M^{\mathrm{fin}}\}. \tag{32}$$

For an incoming annulus, choose a small round circle inside its smooth inner Jordan domain, containing the origin, and fill the complementary annulus by C. The proposed factorization, with the matched central scalar retained, is

$$A_M C_M = \lambda r^{L_0}, \qquad 0 < r < 1, \quad \lambda \neq 0. \tag{33}$$

It puts finite-energy vectors in $\mathrm{Ran}(A_M)$ and gives $C_M M^{\mathrm{fin}} \subset \mathcal{D}_A$. The dense-range property of C_M would make $\mathcal{D}_A$ dense; the adjoint construction treats $\mathcal{D}_{B^*}$. This is the domain issue addressed by the proposed use of Henriques–Tener Lemma 7.9.

Third, Theorem 9.5(i), together with insertion gluing in Proposition 6.19, supplies the incoming identity after the preceding geometric matching. Adjoints supply the outgoing identity. For $\xi \in \mathcal{D}_A$ and $\eta \in \mathcal{D}_{B^*}$, the algebraic pairing

$$\langle Y_M(v, z) A_M \xi, B_M^* \eta \rangle \tag{34}$$

is a finite Laurent polynomial: both vectors have finite energy support, and the mode grading permits only finitely many powers. The geometric insertion pairing is holomorphic on $\mathrm{int}(B, A)$. The proposed geometric argument shows that every connected component meets an incoming or outgoing open part. The identity theorem would then extend equality over the component, including across its seam. Equality on the two dense test domains identifies the bounded extension with $\pi_M(X_v(B, z, A))$. Applying the same reasoning to the vacuum finishes the proposed matching. These details, rather than a general strong-unitarity implication, are the content of Claim A1.

Theorem 10.5 (Conditional complete unitarity)**.** *Assume Claims 10.1, 10.2, and 10.4, together with the preceding algebraic inputs. Then U is completely unitary.*

Proof. Proposition 10.3 unitarizes every simple module. Claim 10.4 supplies the representation required by Tener, Definition 2.18, for each such module; finite direct sums preserve the property. His Corollary 4.6 then supplies positivity of the canonical fusion forms. Strong unitarity must be established first: the unitarizable-fusion-product hypothesis in Theorem 4.4 cannot be used to bypass the exhaustion argument. □

11 Conditional unitary categorical identification

Theorem 11.1 (Unitary Haagerup-center identification, conditional on A1–A3)**.** *Under the hypotheses of Theorem 10.5, and the preceding determination of the all-module fusion ring and its untwisted pointed subcategory, there is a unitary ribbon equivalence*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3).$$

For the original embedded extension H/U, its all-module category is tensor equivalent to one of $\mathcal{H}_2$ and $\mathcal{H}_3$.

Proof. Put $\mathcal{C} = \mathrm{Rep}(U)$ and let A be the algebra of the actual unitary extension H. Complete unitarity allows application of Gui, Theorems 2.21 and 2.28: A is a standard commutative Q-system and its modules are unitarizable. The unitary braiding gives each one-sided module a compatible opposite-side action, realizing $\mathcal{F} = \mathcal{C}_A$ in the unitary bimodule construction. Thus $\mathcal{F}$ is a C^* fusion category. Here $\mathcal{C}_A$ means *all* modules. The local modules form $\mathcal{C}_A^0 \simeq \mathrm{Vec}$ because the extension is holomorphic, so substituting local modules would lose the Haagerup category.

Its already determined fusion ring is $\mathrm{HI}(\mathbb{Z}/3)$. The actual untwisted pointed subcategory makes the obstruction $c^{(0,3)}$ vanish, while the other obstruction is in

$$H^1\Big(\mathbb{Z}/2, H^2(\mathbb{Z}/3, \mathbb{C}^\times)\Big) = 0.$$

These are the hypotheses in Izumi, Definition 2.7 and Theorems 5.9 and 9.1, giving the two unitary tensor classes $\mathcal{H}_2, \mathcal{H}_3$. No algebra structure on $1 + \rho$ is inserted as an extra assumption. Grossman–Snyder, Theorem 1.4, places these categories in the same Morita class. Their centers are therefore braided equivalent. The positive spherical structures make this a ribbon equivalence, and the Lagrangian-center theorem gives

$$\mathcal{C} \simeq Z(\mathcal{F}) \simeq Z(\mathcal{H}_3).$$

Reutter, Theorem 5, provides the unitary braided realization of the equivalence, preserving the associated positive ribbon structures. □

This uses the published unitary classification as an external input; its classification equations have not been replayed here. It avoids the broader pivotal census, but it does not prove that the natural reconstruction functor $\mathcal{R}$ preserves Connes fusion, the VOA tensor product, or braiding. That particular comparison is a distinct statement and is not needed for the theorem just proved conditionally.

12 Conditional full theories and a chosen E_8 extension

Proposition 12.1 (Diagonal full theory; conditional character identification)**.** *The completely rational net $\mathcal{N}_U$ admits its canonical unitary local diagonal full theory. Under Claims 10.1 and 10.2, and the preceding twelve-module character determination, its partition function is*

$$Z_{\mathrm{diag}}(\tau,\bar{\tau}) = \sum_{i=0}^{11} \chi_i(\tau)\overline{\chi_i(\tau)}, \qquad c_L = c_R = 8. \tag{35}$$

Proof. Apply the diagonal Longo–Rehren construction, in the identity-coupling case of Rehren, Theorem 1.4, Proposition 1.5, and Corollary 1.6. Claims A2–A3 identify all sector Hilbert spaces with the twelve ordinary modules and preserve their Hamiltonians. Hence their traces give (35). The annulus compatibility claim is not needed for this net construction or this linear character matching. □

Theorem 12.2 (Chosen $\mathcal{H}_3$ extension and defects; conditional on A1–A3)**.** *Assume the hypotheses of Theorem 11.1 and the preceding strict positivity of all nonvacuum lowest conformal weights. There exists a unitary holomorphic conformal extension $H' \supset U$ isomorphic to V_{E_8} whose all-module category is $\mathcal{H}_3$. For its extension algebra $A_3 \in \mathrm{Rep}(U)$,*

$$\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3, \qquad {}_{A_3}\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}. \tag{36}$$

The second category describes algebra-bimodule defects preserving the U chiral subalgebra in the corresponding full extension theory and contains an $\mathcal{H}_3$ fusion subcategory.

Proof. Let $I_{\mathcal{H}_3}$ be the right adjoint of the forgetful functor $Z(\mathcal{H}_3) \to \mathcal{H}_3$. Transport the canonical algebra $I_{\mathcal{H}_3}(\mathbf{1})$ along the unitary ribbon equivalence. By the canonical Lagrangian-algebra construction, as in Davydov–Müger–Nikshych–Ostrik, Section 4, the resulting A_3 is connected, commutative, separable, and twist-free, with

$$\mathrm{FPdim}(A_3)^2 = \mathrm{FPdim}(\mathrm{Rep}(U)), \qquad \mathrm{Rep}(U)^0_{A_3} \simeq \mathrm{Vec}, \qquad \mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3.$$

Trivial twist makes its constituent conformal weights integral. Connectedness and strict positivity away from the vacuum give a one-dimensional weight-zero space. The extension correspondence of Huang–Kirillov–Lepowsky, Theorems 3.2–3.6, together with Gui's unitary Q-system theorem, produces a unitary conformal VOA extension H'. It is a finite sum of ordinary U-modules; finite-extension C_2-cofiniteness and the semisimple local-module description give strong rationality. The Lagrangian condition makes it holomorphic. Its central charge is eight, so Dong–Mason, Theorem 1, identifies H' with V_{E_8}.

For the canonical algebra, the algebra-bimodule duality gives the second equivalence in (36); see Müger, Sections 4 and 8. Here "op" reverses the tensor product. The first-factor embedding $x \mapsto x \boxtimes \mathbf{1}$ supplies the stated $\mathcal{H}_3$ fusion subcategory. □

Remark 12.3 (Extension choice and preserved symmetry)**.** The selection of the canonical $\mathcal{H}_3$ algebra proves existence of H' with the specified all-module category. It does not identify A_3

with the original embedded H/U algebra; the original boundary category may be $\mathcal{H}_2$ or $\mathcal{H}_3$. Both holomorphic VOAs being isomorphic to V_{E_8} does not identify their embeddings of U. Moreover, the defects in (36) preserve U and need not preserve the enlarged E_8 chiral algebra. The local chiral H'-module category is Vec.

The logical separation is therefore explicit: completed finite certificates prove the actual strong-generation and properness result; the algebraic identification supplies its stated representation-theory inputs; A2–A3 supply sector exhaustion and strong unitarity; A1 supplies the additional analytic comparison needed for complete unitarity. The unitary categorical and $\mathcal{H}_3$ extension conclusions retain those conditions until the highlighted applications are independently verified. The synthesis sources are `hi_cft_20260923_c8two_final_theorem.md` and `hi_cft_20260923_c8two_certificate_audit.md`; their unqualified analytic conclusion is deliberately recorded here with the review boundary made visible.

13 Comparison with arXiv:2512.08225

Albert–Honda–Kaidi–Zheng explicitly assume the existence of a $c = 8$ Haagerup-center theory in Section II, footnote 4 of arXiv:2512.08225v2. Their equation (12) displays the γ-dependent characters, and their extension discussion relates the center theory to the E_8 theory with Haagerup defects.

If the highlighted claims of this manuscript are confirmed, the explicit algebra (1) supplies their $\gamma = 0$ existence premise. The choice of an $\mathcal{H}_3$ Lagrangian algebra supplies a compatible E_8 extension, with the distinction between U-preserving and full-E_8-preserving defects maintained. Neither this construction nor that finite gauging identifies the continuum limit of the ρ-projector chain. Section III of the cited paper proposes further asymmetric quotients towards charges two and six; those require additional conformal and defect data. No uniqueness theorem for all central-charge-eight CFTs is asserted here.

A Certificate ledger and reproducibility

The following are existing, closed records. No new lattice computation was run while preparing this typeset manuscript. All paths are relative to the project directory. The long filenames deliberately preserve an unambiguous connection to the actual records.

Result	Record and scope
Lower identities	`hi_cft_20260923_next_structure_lower_closure.json`: 3,970 exact relations.
Actual triplet	`hi_cft_20260923_next_structure_p4_controls.json`: 18 current-zero and six direct Virasoro controls.
Raising saturation	`hi_cft_20260923_c8two_structure_saturation.json`: 644 exact images in a quotient of dimension 322.
Additional direct seeds	`hi_cft_20260923_c8two_targeted_h4_trimmed_v2/summary.json`: 118 completed exact identities. The unfinished final block in this historical directory is not counted as a completed identity.
Raised seeds	`hi_cft_20260923_c8two_structure_h4_raised_exact/summary.json`: twelve additional ordinary exact identities.

Result	Record and scope
Singlet premise	`hi_cft_20260923_c8two_h4_singlet_premise_verified.json`: all 24 raising conditions replayed.
Projected residuals	`hi_cft_20260923_c8two_h4_singlet_exact.json`: twelve full exact residuals, using 21 selected columns.
Full Gate 4	`hi_cft_20260923_c8two_gate4_verified.json`: all 2,898 identities, using singlet recovery and raising saturation.
Charged spanning	`hi_cft_20260923_c8two_charged_span_verified.json`: 28 reconstructed seeds, rank 322, and current commutators.
Full Gate 5	`hi_cft_20260923_c8two_gate5_verified.json`: all 24 channels by the stated Jacobi argument.
Combined provenance	`hi_cft_20260923_c8two_proof_bundle.json`: 640 authenticated files, including earlier exact inputs and proof notes. It checks provenance and completion, not formal correctness of every mathematical argument.

The final Gate 4 verifier replays the twelve projected identities using integer-cleared coefficients, separately from the rational producer. A field element is represented in the basis $1, s, t, st$, with $s^2 = 13$, $t^2 = -3$; all residuals are exact four-component rational identities. Prime calculations are used to select candidate columns and to certify positive spanning against an independently known upper dimension. A vanishing prime residual alone is never treated as a characteristic-zero containment proof.

The provenance-ledger replay can be written to a fresh output without changing the original certificate:

```
python hi_cft_safe_run.py --limit-mib 1024 -- \
  python hi_cft_20260923_c8two_proof_bundle.py \
  --output /tmp/haagerup-c8-proof-review.json
```

These lines represent a single shell command. The output path must not already exist. This rechecks the frozen source hashes and completion flags; it does not rerun all original mode calculations. The source files of the individual certificate producers and verifiers are bound into the same ledger.

B Compilation and review scope

The main source is `hi_cft_c8_proof.tex`; its companion sources are `hi_cft_c8_manuscript_structure.tex` and `hi_cft_c8_manuscript_analytic.tex`. Run `pdflatex` twice on the main source for cross-references and the table of contents. The document uses standard TeX packages and requires no shell escape.

The computational records predate this manuscript and remain unchanged. The original provenance ledger consequently does not authenticate this new exposition. The PDF organizes and qualifies those records; it is not itself an additional mathematical certificate. Before treating the full identification as externally established, the highlighted local claims should be checked by specialists and the certificate recipes should be replayed in an independently reviewed environment.